

## 31338 Network Servers

### Lab 3a: Static Networking

#### Aims

1. Use command-line tools to configure static networking
2. Make persistent network settings (for static networking) by editing networking configuration files

#### Assumed Knowledge

Basic IP networking: IP addresses, Subnets, Routing

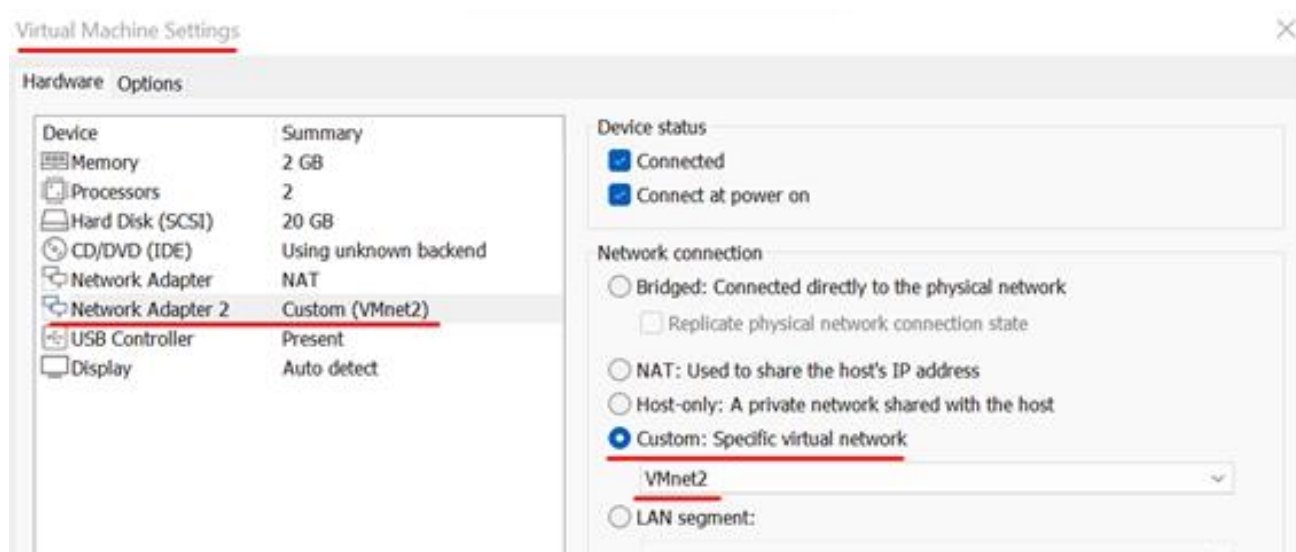
#### Background Information

We will set up a network initially using static addressing and will test our servers by pinging and then using ssh to each other.

#### Task 1: Querying your network configuration

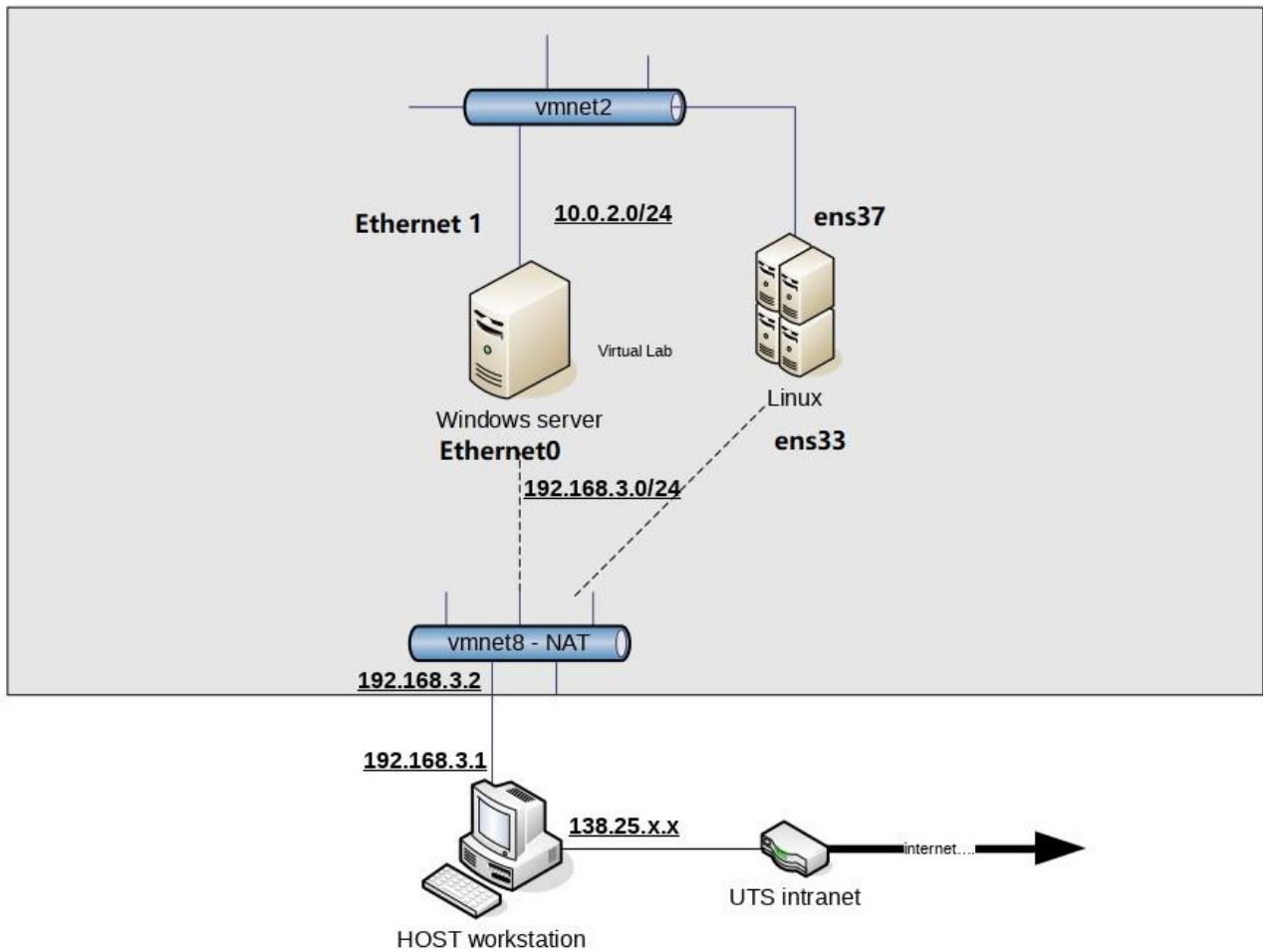
We assume that you are aware of the "dotted quad" conventions. We will use the "virtual switch" vmnet2 as our new private network.

Go to “Settings” of your virtual machine to change the Network Adapter to “Custom” using VMnet2.



Take a few minutes to look at the above picture and compare this to your own configuration.

### Static Network Lab



Use the following commands to check your EXISTING configuration.

Hint: On Linux, make sure the network is up, i.e. use systemctl enable NetworkManager.service

| Linux                | Windows                    |
|----------------------|----------------------------|
| ifconfig<br>route -n | ipconfig<br>route print -4 |

Enter the default values here for the record:

|               |  |                    |  |
|---------------|--|--------------------|--|
| Linux: ens33  |  | Windows: Ethernet0 |  |
| DNS suffix    |  | DNS suffix         |  |
| Inet address  |  | Inet address       |  |
| Subnet mask   |  | Subnet mask        |  |
| Default route |  | Default route      |  |

HINT: There are also graphical user interfaces to the network management tools. (Linux) Settings -- Network  
(Windows Server) Server Manager -- View Network Connections; then double-click on the interface and choose “details”

The command to check the status of all network interfaces on Linux is: `ifconfig -a` (/sbin/ifconfig). Global network configuration parameters (i.e. not interface specific) are in /etc/sysconfig/network Interface-specific network parameters are in /etc/sysconfig/network-scripts/ifcfg-ens33 for the Ethernet 0 interface and in /etc/sysconfig/network-scripts/ifcfg-ens37 for Ethernet 1 interface. In older machines you may find these named ‘eth0’ and ‘eth1’, but current machines use the [Consistent Network Device Naming](#) conventions.

The virtual network between your virtual machines and the host workstation is probably set to 192.168.3.0/24 if you are in the lab at UTS. If you are at home, it may be different depending on your VMware configuration of “VMnet8” virtual network.

Note: please confirm this – it can vary on your original host workstation. In the description below, if your network is different, please translate all references to 192.168.3.0/24 to your network ...

On this virtual network, IP addresses from 192.168.3.128 to 192.168.3.254 are allocated by DHCP to both the Ethernet 0 interfaces on the Linux and Windows VM.

Notice that we don’t have a network connection for the second adapter – ens37 on Linux and Local Area Connection 2 on Windows Server.

Note: Windows uses Automatic Private IP Addressing (APIPA) to allocate an address in the 169.254.0.0/16 address range. We will change this in the next task.

## **Task 2: Designing the network**

We are going to set up our own private network on the 2<sup>nd</sup> network adapter via the VMnet2

virtual switch. Try using your IP skills to decide on the following setup configuration:

1. Our network will be in the subject 10.0.2.0/24
2. We will have to eventually cope with 200 machines
3. Our gateway will be the Linux machine
4. Pick an appropriate IP address/netmask/gateway for the Linux and Windows servers

Calculate the appropriate subnet and host values here:

| Linux: ens37  |               | Windows: Local Area Connection 2 |               |
|---------------|---------------|----------------------------------|---------------|
| Inet address  | 10.0.2.1      | Inet address                     | 10.0.2.2      |
| Subnet mask   | 255.255.255.0 | Subnet mask                      | 255.255.255.0 |
| Default route | 10.0.2.1      | Default route                    | 10.0.2.1      |

### **Task 3: Command-line configuration**

#### **3a: Linux server**

Initially, your machine is configured to use DHCP networking. But for now, we want to set the IP address manually, using command-line tools. Use the `ifconfig` command to manually configure the network parameters for Ethernet 1 (ens37). Use the above calculated values for your system.  
e.g.

```
ifconfig ens37 [your-ip-addr] netmask [your-netmask]
```

The netmask option may be omitted, in which case the netmask will default to the classful netmask (which is not appropriate here since we have a private class A address (recall 10.0.0.0/8 ?))

View the routing table using the `route` command (or `netstat -r`). Next, add a default route that will send traffic to the virtual network's default gateway, which is shown below. Although the default gateway is already set to the correct IP address for your topology, it is important to know and be able to use the command for setting a default gateway.

```
route add default gw [gateway-ip-addr]
```

Now, test that your statically configured network is working by trying to `ping [gateway-ip-addr]` (which you defined earlier).

Can you try pinging your Linux server from your host workstation?

Try pinging both interfaces: the original ens33, then the ens37 address you allocated. Which one works? Can you guess why you mightn't be able to ping the other?

#### **3b: Windows server**

On Windows Server, we will use the GUI to configure our 2nd network card.

- Server manager --- Local Server --- then click on the interface (Ethernet1) to bring up the Network Connections control panel
- Right-click on the interface (Ethernet1) --- Properties
- Choose "Internet Protocol Version 4 (TCP/IPv4)" --- Properties
- Then choose the "Use the following IP Address" & enter the values you calculated.

Confirm that the configuration is correct (recall the `ipconfig` command ?)

Now you should be able to ping the Linux server

Verify that this is two way by pinging from the Linux server to your Windows server.

If you can't, you may need to set up your firewalls to enable the ping command. See the next optional task to set up the firewall on both Linux and Windows.

## **Task 4: Setting up the firewall**

You probably found that you could ping from Windows to Linux but not from Linux to Windows. This is because by default, Windows Server disables the ping (ICMP Echo) response.

### **4a: Windows server**

First, be aware that Windows runs all network adapters in various “profiles” for each network location. These profiles are usually: Private, Domain, and Public (with increasing levels of paranoia)

There are 2 ways to modify the Windows firewall:

1. (easiest) Click Start --- Control Panel --- Network and Internet --- System and Security- >Windows Defender Firewall --- Allow an app through firewall --- File and Print sharing --- enable checkbox(es) (private and public) --- Press OK button.
2. (advanced) Server manager --- Tools --- Windows Defender Firewall with Advanced Security --- Inbound rules--- File and Printer sharing (Echo request ICMPv4 IN, domain....) --- enable for each profile

As you can see, you can do fine grain firewall management through this management console.

Note that if you choose option 1 (Control Panel --- Windows Firewall), then look at the Server Manager, this will enable all the File and Print sharing options. BE VERY CAREFUL WITH THIS OPTION – you are also opening up the file sharing ports on your windows firewall (Printing, SMB, RPC) not just the ICMP (ping) functionality.

### **4b: Linux**

Different versions of Linux can use different firewall packages. The package used by Centos is ‘firewalld’. Verify that firewalld is active (currently running) and enabled (will start at boot) with `systemctl status firewalld`

Linux is considerably harder to set up, so we will install the firewall-configpackage. Do this

now! Run the command with: `firewall-config`

In the connections list on the left, check the zone that your ens37 interface is in (usually “public”). Then in the main part of the window, make sure you are in the ‘Zones’ tab and the public zone is selected. There is a set of sub-tabs, e.g. Services, Ports, Protocols, etc. If you scroll across you should find a tab called “ICMP Filter”. By default the boxes are unchecked. Unlike Windows, Linux by default allows all ping (ICMP Echo) requests.

If you turn on the checkbox for echo-reply, Linux should REJECT the incoming pings.

Apply this, and then check that you now can’tping from Windows.

Finish off this task by re-enabling ping (ICMP reply).

The command-line equivalents of this are:

- `firewall-cmd --add-icmp-block=echo-reply`
- `firewall-cmd --add icmp-block=echo-reply`
- `firewall-cmd --query-icmp-block=echo-reply`

## **Task 5: Modifying Linux network configuration files**

The statically configured Linux network setup described above will, of course, vanish on reboot and be replaced by whatever is specified in the startup network configuration files. The next task involves modifying the network configuration files.

Unfortunately this is quite different for different varieties of Linux. Here we will work with the Fedora/Redhat/Centos approach (of course), but unlike the steps described in Task 1, most of this knowledge is not transferable to other varieties of UNIX/Linux.

On Centos, the two main configuration files to configure static networking are /etc/sysconfig/network (global parameters that apply to all interfaces) and /etc/sysconfig/network-scripts/ifcfg-ensXX (interface-specific parameters for Ethernet interfaces).

First, edit/view ifcfg-ens33(Ethernet 0). BOOTPROTO should say BOOTPROTO=dhcp, meaning that this interface will be dynamically assigned an IP address by a DHCP server. Because of this there shouldn't be an IPADDR or NETMASK entry in this file. Also note/check that DEFROUTE=yes. This means that this interface should be the default route/gateway. Finally, look for ONBOOT. If it shows ONBOOT=no then change it to ONBOOT=yes.

Then edit ifcfg-ens37(or create it if it doesn't exist by copying ifcfg-ens33):

- Change the line that starts with DEVICE= to say DEVICE=ens37
- Change the line that starts with NAME= to say NAME=ens37
- If there is a UUID line, delete it
- Because our ens37 interface should have a static IP address, next change the BOOTPROTO line to say BOOTPROTO=none
- Then set the variables IPADDR and NETMASK to the same values as you used for Task 1 (e.g. Add the line IPADDR=10.0.2.x where x was your host TCP/IP address)
- Make sure it says DEFROUTE=no (this is not the default route)
- Also make sure this interface will start up automatically when the machine boots (ONBOOT=yes)

When you are ready to test your configuration, use the ifdown and ifup scripts in the network-scripts directory. Bring the interface down first, and then up:

```
ifdown ens37  
ifup ens37
```

Alternatively to restart/reload all networking, you can run: `systemctl restart NetworkManager.service`

A third way is to use the NetworkManager command nmcli. Increasingly this is the preferred way to manage network connections rather than ifconfig/ifdown/ifup. Try the following commands and compare them with ifconfig/ifdown/ifup. Note similarities/differences in your journal.

- nmcli con show ens37
- nmcli con down ens37
- nmcli con up ens37

A fourth way is to use the command nmtui. Try it.

After making changes and bringing the interfaces down/up, use the ifconfig and route commands to check that the configuration looks correct. Test by using ping to and from the two servers.

Reboot your main VM and test that all of the changes you made are persistent.

HINT: We have shown you the command line version of updating a Centos Linux server. Of course the easiest way is to use the graphical interface, which updates these very same files.

Make sure you have documented the names, paths and contents of these configuration files in your journal. This lab is important, as many of the exercises in later labs involve setting up and testing servers, and so rely on a fully operational network.